

PROBABILITY: SAMPLE SPACE & TREE DIAGRAMS

Fun with Chances and
Choices!





What is Probability?

Probability tells us how likely something is to happen!

Everyday examples:

- Tossing a coin – Will it be heads or tails?
- Rolling dice – Which number will come up?
- Picking a card – What color will you get?

Probability helps us guess what might happen next!

Let's Explore!

Understanding Sample Space

Sample space is the list of ALL possible outcomes.

Example: Tossing a coin $\rightarrow$ {Heads, Tails}



Sample Space: Dice Roll

Outcomes: {1, 2, 3, 4, 5,
6}



Sample Space: Tossing Two Coins

HH (Heads, Heads)

HT (Heads, Tails) • TH
(Tails, Heads)

TT (Tails, Tails) — 4
outcomes total!



Try It Yourself!

What's the Sample
Space?





What is a Tree Diagram?

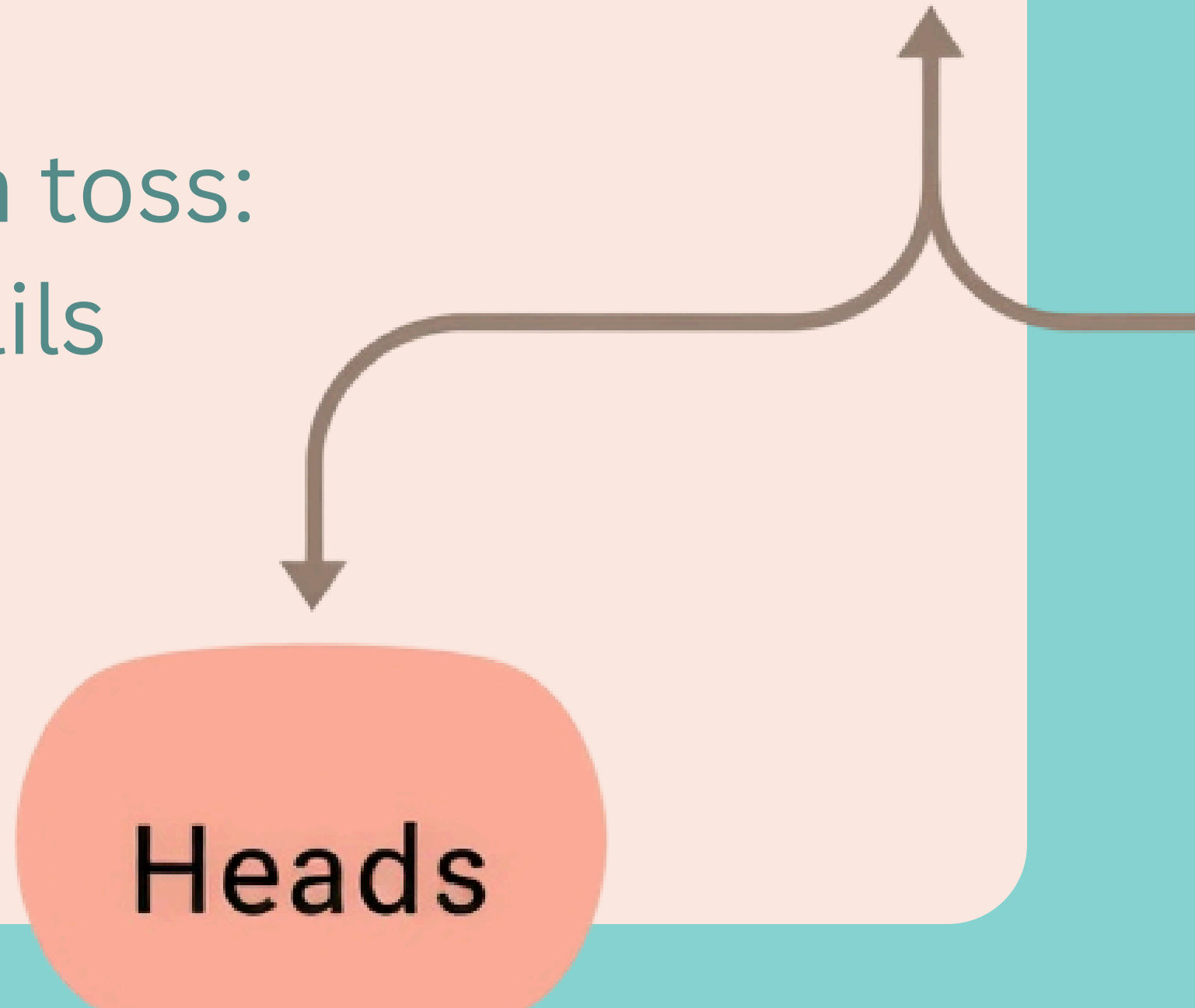
A tree diagram helps us list all possible outcomes step by step!

Why use it? To organize outcomes clearly and see all choices at once. Think of it like branches on a tree — each branch shows a different choice or result!

Let's
Organize!

Building a Tree Diagram: Step 1

Start with one coin toss:
Heads $\rightarrow$ or $\leftarrow$ Tails



Heads

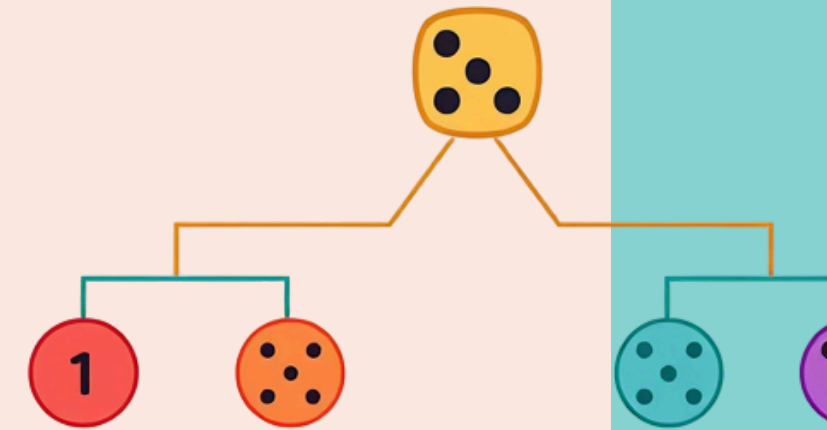
Building a Tree Diagram: Step 2

HH (Heads,
Heads)
HT (Heads, Tails)
TH (Tails, Heads)



Tree Diagram: Dice & Coin

$2 \times 2 = 4$
outcomes!



Using Tree Diagrams to Find Probability

Favorable Outcomes $\div$
Total Outcomes



Practice Problem 1

If you toss a coin and roll a dice,
what is the chance you get Tails and
a 4?

$$? 2 + = ?$$



Practice Problem 2

$P(\text{exactly 1 Head}) =$
?



Summary: What We Learned Today



Probability means chance



Sample space is all possible outcomes



Tree diagrams help list outcomes stepwise





Keep Exploring Probability!

Probability is all around us! Keep asking "What can happen?" Try fun experiments at home like coin toss games, dice rolls, or picking cards. The more you explore, the better you'll understand chances and choices. Math is everywhere!

Thank You for Learning with
Us!

